Day 153: Week 22 Build: a Prophet-Style Forecaster from This Week's Parts

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 22, Day 7 | Complete study guide

Watch: Sean Taylor - Forecasting at Scale: How and Why We Developed Prophet for Forecasting at Facebook | Forecasting with the FB Prophet Model (Rob Mulla)

Script: `day153_prophet_style_forecaster.py` · every number in this guide is copied by program from `day153_run_output.txt`, the real output of that script.

A trend that may bend at 25 changepoints plus a season of Fourier waves, fitted by one penalised least squares, with Prophet's simulated intervals. It describes CO2 well: its season matches Day 147's within 0.028 ppm, and its slope rises from 0.745 to 1.546 ppm a year. As a forecaster it was middling on 2000–2001 (0.400 ppm, against the airline ARIMA's 0.412 and the control's 0.231), lost to the airline model at every horizon over 97 rolling origins, and its 80% intervals held only 0.536 to 0.289 of the time. The reason was measured: what it leaves behind has strong memory (lag-1 autocorrelation 0.819), which the curve ignores.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

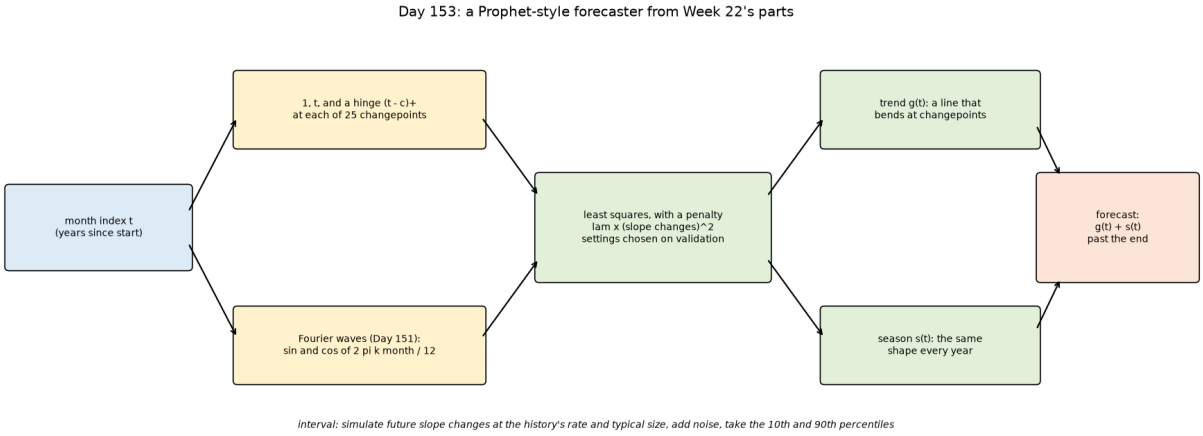
A **decomposable** forecaster writes the series as a sum of parts that are each a function of the date: a **trend** g(t), a **season** s(t), and an error. Prophet (Taylor and Letham, 2017) made this popular. Its trend is a straight line whose slope may change at many candidate months, the **changepoints**; a penalty keeps most changes small. Its season is a sum of sine and cosine waves (Day 151's Fourier columns).

Because every part is linear in its coefficients, the whole model is one least-squares fit. The forecast is the same curve carried past the last month: the last slope continues, and the season repeats.

REAL-WORLD SCENARIO

An analyst has hundreds of business series and no time to fit an ARIMA to each. A curve-fitting tool that needs no differencing, copes with missing months and draws its own trend and season is attractive. The question for today is what it gives up.

SYSTEM DIAGRAM



The forecaster: hinge and wave columns, one penalised least squares, trend plus season.

Built from the week's parts, imported unchanged: Day 147's data and classical decomposition, Day 148's autocorrelation and KPSS, Day 149's airline ARIMA, psi weights and Ljung-Box test, Day 150's Holt-Winters, Day 151's Fourier columns, least squares and regression, and Day 152's LSTM and control. Today's build has no exercise file: it is the exercise files, imported. Prophet itself puts a Laplace prior on the slope changes and fits with Stan; here the penalty is squared (ridge), so the fit stays one call to least squares. CO2 is trained up to 1999-12 and tested on 2000–2001; every setting is chosen on validation origins whose forecasts end by 1989-12.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. The parts, checked

DEFINITION — start here, no prior knowledge assumed

Each changepoint c adds a **hinge** column $(t - c)_+$: zero before c, then rising by one a year. Its coefficient is the change in slope at c. With 25 hinges, the trend can bend 25 times; the penalty decides how much. The candidates are spread evenly over the first 80% of the history, as in Prophet, so the last stretch is a single straight piece whose slope the forecast carries on.

REAL-WORLD SCENARIO

A road surveyor fits a route with straight stretches joined at bends, and is charged for every degree of bend: only the bends that really help are taken.

MATHEMATICS

$$y(t) = a + k\,t + \Sigma_j \delta_j\,(t - c_j)_+ + \Sigma_{k=1..K}\,[\,a_k \sin(2\pi k\,m / 12) + b_k \cos(2\pi k\,m / 12)\,] + e \quad (m = \text{calendar month})$$

$$\text{minimise} \quad |y - X\beta|^2 + \lambda\,\Sigma_j\,\delta_j^{\,2} \quad = \quad \text{least squares on } X \text{ stacked over } \sqrt{\lambda} \text{ rows, one per slope change}$$

$$\text{slope after the last changepoint} = k + \Sigma_j\,\delta_j$$

IMPLEMENTATION

```
def changepoints(n, count=25, frac=0.8):
    return np.unique(np.linspace(0, int(frac * n), count + 1).round().astype(int)[1:])
def design(start, idx, cps, K):
    t = idx / 12.0
    H = np.maximum(t[:, None] - cps[None, :] / 12.0, 0.0)
    return np.column_stack([np.ones(len(idx)), t, H, d151.fourier(np.datetime64(start, "M") + idx,
        K)])
def fit_prophet(y, start, K=4, lam=100.0, count=25, frac=0.8):
    n = len(y); cps = changepoints(n, count, frac) if count else np.array([], int)
    X = design(start, np.arange(n), cps, K)
    P = np.zeros((len(cps), X.shape[1])); P[:, 2:2 + len(cps)] = np.sqrt(lam) * np.eye(len(cps))
    beta = np.linalg.lstsq(np.vstack([X, P]), np.r_[y, np.zeros(len(cps))], rcond=None)[0]
    r = y - X @ beta
    return dict(beta=beta, cps=cps, K=K, lam=lam, start=start, n=n, sigma=r.std(), resid=r,
        delta=beta[2:2 + len(cps)])
```

ACTUAL OUTPUT

```
a line with slopes 1, 3, 0.5 a year (changes at months 60 and 120), a season and noise sd 0.3; changepoints placed at [60,
    120]
  fitted slopes 0.937, 3.058, 0.449 a year; noise sd left over 0.287
CO2 to 1999-12, no penalty: against Day 151's fit_ols on the same columns, largest coefficient difference 2.1e-13
penalty 100: stacked least squares against the closed form (X'X + lam D) b = X'y, largest difference 1.9e-11
no changepoints: fitted values against Day 151's regression with a trend and 4 waves, largest difference 2.3e-13
the forecast's trend rises 1.5456 ppm a year; the last piece's slope is 1.5456
1000 simulated futures, 24 months each; their mean against the forecast, largest gap 0.037 ppm (noise sd 0.525)
Actual output from this lesson's run.
```

What it means

On a made-up series with slopes 1, 3 and 0.5 a year, the fit recovered 0.937, 3.058 and 0.449, leaving noise of sd 0.287 (the true 0.3). With no penalty, the coefficients equal Day 151's fit_ols on the same columns to 2.1e-13; with a penalty, the stacked least squares equals the closed form to 1.9e-11; with no changepoints, the fit equals Day 151's regression with a trend and four waves to 2.3e-13.

The forecast's trend rises 1.5456 ppm a year, exactly the last piece's slope, and 1000 simulated futures average to the forecast within 0.037 ppm, small against the noise sd of 0.525.

WHY CHOOSE THIS?

One least-squares solve: fast, exact and easy to check, with every part readable.

WHY NOT THIS? (TRADEOFFS)

The trend is only ever straight pieces, and the last piece's slope is carried on unchanged, however long the horizon.

2. Choosing the waves and the penalty

DEFINITION — start here, no prior knowledge assumed

Two settings matter: K, the number of wave pairs, and λ, the penalty on slope changes. Each pair was tried at 97 origins from 1980-01 to 1988-01, refitting the curve and forecasting 24 months each time; every forecast ends by 1989-12. After choosing, the test error of each penalty is shown, only to see whether the choice held.

REAL-WORLD SCENARIO

Tuning a recipe on last year's harvest before cooking with this year's: the test is whether last year was a good guide.

IMPLEMENTATION

```
def choose(y, start, origins, h=24, Ks=KS, lams=LAMS):
    E = np.array([[np.mean([np.mean(np.abs(prophet_forecast(fit_prophet(y[:o], start, K, lam), h)
        - y[o:o + h])) for o in origins])
                   for lam in lams] for K in Ks])
    i, j = np.unravel_index(np.argmin(E), E.shape)
    return Ks[i], lams[j], E
```

ACTUAL OUTPUT

```
97 origins, 1980-01 to 1988-01 (1.0 s); mean MAE (ppm) over 24 months
   waves  pen 0.01   pen 0.1     pen 1    pen 10   pen 100  pen 1000
       2     0.495     0.497     0.500     0.491     0.481     0.557
       3     0.489     0.491     0.494     0.485     0.476     0.554
       4     0.489     0.491     0.494     0.485     0.474     0.551
       5     0.489     0.490     0.494     0.485     0.474     0.551
validation's choice: 5 waves, penalty 100 (0.474); no changepoints at all (a straight line) 2.606
did the choice hold? test MAE (2000-2001) with 5 waves, by penalty: 0.01: 0.235, 0.1: 0.235, 1: 0.232, 10: 0.255, 100:
    0.400, 1000: 0.447
  the choice ranks 5 of 6 on the test; the test is reported, not used to choose
Actual output from this lesson's run.
```

What it means

Validation chose 5 waves, penalty 100 (0.474); the table is flat across waves, and only a heavy penalty of 1000 or a trend with no bends at all (2.606) clearly lost.

The choice did not hold: on 2000–2001 it ranked 5 of 6. A light penalty scored 0.232, the chosen one 0.400. Why the 1980s favoured a stiff trend and 2000–2001 a flexible one was not tested here; picking the light penalty now would be choosing on the test, so the chosen setting stays.

WHY CHOOSE THIS?

Time-ordered validation chooses without touching the test, the same rule as all week.

WHY NOT THIS? (TRADEOFFS)

With a flat validation table, the winner is close to a coin toss, and it ranked fifth of six later.

3. What the model says about CO2

DEFINITION — start here, no prior knowledge assumed

Fitted to 1999-12, the curve can be read directly: the slope in each piece, the season by calendar month, and the residuals. The season is compared with Day 147's classical decomposition, and the residuals are tested with Day 148's autocorrelation and KPSS and Day 149's Ljung-Box test.

REAL-WORLD SCENARIO

A doctor's growth chart: a smooth curve for the trend, a pattern for the time of year, and the question of whether the points that miss it are random or keep missing in the same direction.

IMPLEMENTATION

```
def components(F, idx):
    X = design(F["start"], idx, F["cps"], F["K"]); m = 2 + len(F["cps"])
    return X[:, :m] @ F["beta"][:m], X[:, m:] @ F["beta"][m:]
def prophet_forecast(F, h):
    return sum(components(F, np.arange(F["n"], F["n"] + h)))
```

ACTUAL OUTPUT

```
25 changepoints, 1959-07 to 1991-08; the slope went from 0.745 ppm a year (1958) to 1.546 (after 1991-08)
  largest slope changes: 1991-08 +0.147, 1988-12 -0.125, 1976-12 +0.107
the season (5 waves) against Day 147's classical decomposition: largest difference 0.028 ppm (peak 2.92 against 2.93, month
    5 of 12)
what is left: sd 0.525 ppm; autocorrelation at lags 1, 2, 12: 0.819, 0.712, 0.317
  Ljung-Box over 24 lags: Q = 1633.6 (5% critical value about 36); the airline model's residuals Q = 20.6
  KPSS on what is left: 0.038 (5% critical value 0.463): it wanders, but around a fixed level
the curve captures the trend and the season; the month-to-month memory is left in the errors, and it forecasts
as if it were not there
Actual output from this lesson's run.
```

What it means

The slope rose from 0.745 ppm a year in 1958 to 1.546 after 1991-08; the largest single change was +0.147 there. The season matches Day 147's to 0.028 ppm, peaking at 2.92 in May.

What is left is far from noise: its autocorrelation is 0.819 at lag 1, and the Ljung-Box statistic is 1633.6 against about 36 for noise (the airline model's residuals: 20.6). KPSS (0.038) says it does not drift away: the curve misses in long runs, too high for many months, then too low. The forecast treats those runs as if they did not exist.

WHY CHOOSE THIS?

Interpretable: a slope in ppm a year and a seasonal shape anyone can read off a plot.

WHY NOT THIS? (TRADEOFFS)

Months are treated as independent around the curve, which these residuals are not.

4. Forecasting CO2 in 2000–2001, against every model of the week

DEFINITION — start here, no prior knowledge assumed

All models are fitted on data to 1999-12 and forecast the same 24 months. Two variants test today's findings: the curve with **no changepoints**, and the curve **plus an AR(1) on what it leaves behind**, which adds the last residual, shrinking by ϕ each month. If the residuals' memory is what the curve misses, the AR(1) should help most at short horizons. The interval follows Prophet: future slope changes are simulated at the history's rate, with sizes drawn from a Laplace distribution at the history's average size, plus noise.

REAL-WORLD SCENARIO

A weather forecaster who knows the climate for the date, and one who also looks out of the window this morning.

MATHEMATICS

AR(1) correction: $\phi = \Sigma\, r_{t-1}\, r_t \,/\, \Sigma\, r_{t-1}^{\,2}$, $\mathit{forecast}_{T+h} = \mathit{curve}_{T+h} + \phi^h\, r_T$

interval: future month j gets a slope change with probability 25 / n, size ~ Laplace(0, mean |δ|); 10th and 90th percentiles of 1000 futures

IMPLEMENTATION

```
def prophet_sims(F, h, sims=1000, seed=0):
    r = np.random.RandomState(seed)
    rate = len(F["cps"]) / F["n"]
    scale = np.mean(np.abs(F["delta"])) if len(F["delta"]) else 0.0
    size = r.laplace(0, scale, (sims, h)) * (r.rand(sims, h) < rate)
    ramp = (np.cumsum(np.cumsum(size, axis=1), axis=1) - np.cumsum(size, axis=1)) / 12.0
    return prophet_forecast(F, h) + ramp + r.randn(sims, h) * F["sigma"]
def ar1_correction(F, h):
    r = F["resid"]; phi = (r[:-1] @ r[1:]) / (r[:-1] @ r[:-1])
    return r[-1] * phi ** np.arange(1, h + 1), phi
```

ACTUAL OUTPUT

```
all models fitted on data to 1999-12 (75.0 s, mostly the networks); AR(1) on the residuals: phi = 0.822
            2000-01 to 2001-12, 24 months     MAE    bias
      Prophet-style (validation's choice)   0.400  -0.399
    Prophet-style + AR(1) on what is left   0.295  -0.261
  straight line + season (no changepoints)   3.062  -3.062
                  airline ARIMA (Day 149)   0.412  +0.405
            Holt-Winters, mult. (Day 150)   0.329  +0.314
          regression, lags 1-13 (Day 151)   0.706  +0.706
             LSTM, average of 5 (Day 152)   0.250  +0.196
        control: average change (Day 152)   0.231  +0.066
           same month last year (Day 147)   1.882  -1.882
Prophet-style 80% interval: half-width 0.68 ppm one month ahead, 0.66 at 24; 20 of 24 months inside
Actual output from this lesson's run.
```

What it means

The curve scored 0.400, about level with the airline ARIMA (0.412) but behind Holt-Winters (0.329), the LSTM (0.250) and the control (0.231). It forecast too low by -0.399 on average. Without changepoints it scored 3.062: one straight line since 1958 cannot follow CO2's quickening rise.

The AR(1) on its residuals ($\phi = 0.822$) cut the error to 0.295. The curve's own 80% interval held 20 of 24 months, but it barely widens: half-width 0.68 one month ahead and 0.66 at 24. A likely reason, not tested here: the heavy penalty keeps the fitted slope changes small, so the simulated future ones are small too.

WHY CHOOSE THIS?

The changepoints carry the model: without them its error was nearly eight times larger here.

WHY NOT THIS? (TRADEOFFS)

One test window is a small sample; the rolling origins below are the stronger evidence.

5. Rolling origins, and how often the intervals held

DEFINITION — start here, no prior knowledge assumed

97 origins from 1990 to 1998, as on Days 150–152. The curve is one least-squares solve, so it is refitted at every origin, and so is the control; a version of the curve fitted once in 1989 shows why that matters. The other models are fitted once on data to 1989-12 and read the history up to each origin, as on Day 152. For each origin and horizon, the check is whether the actual month fell inside the 80% interval: Prophet-style's simulated one, and the airline model's from Day 149's psi weights.

REAL-WORLD SCENARIO

Testing a weather service's "80% chance" claims over many days: if it is right, rain falls inside its range on about 80% of them.

ACTUAL OUTPUT

```
97 origins (60.4 s); the curve and the control recomputed at every origin, the rest fitted once on data to 1989-12
       MAE (ppm), months ahead       1       6      12      24
       Prophet-style, refitted   0.631   0.766   0.942   1.257
    Prophet-style, fitted once   1.411   1.443   1.453   1.474
         Prophet-style + AR(1)   0.294   0.696   0.925   1.256
                airline ARIMA   0.264   0.453   0.717   1.101
          Holt-Winters, mult.   0.277   0.459   0.684   1.036
         regression, lags 1-13   0.314   0.650   0.736   1.127
            LSTM, average of 5   0.252   0.534   0.630   0.949
       control: average change   0.273   0.600   0.680   1.069
the refitted curve beats the airline model at 0 of 24 horizons; with AR(1) added, at 0 of 24
       inside the 80% interval       1       6      12      24
                Prophet-style   0.536   0.443   0.361   0.289
                airline ARIMA   0.845   0.753   0.732   0.701
Actual output from this lesson's run.
```

What it means

Refitted, the curve missed by 0.631 ppm one month ahead, against the airline model's 0.264, and beat the airline model at 0 of 24 horizons. Fitted once, it was worse still (1.411): a curve of time cannot use months it was not fitted on. With the AR(1) on residuals, the one-month error fell to 0.294, but by 24 months the gain was gone (1.256 against 1.257), as a decaying correction must be. The LSTM was best at long horizons (0.949 at 24).

The intervals promised 80% and held 0.536 of the time one month ahead and 0.289 at 24. The airline model's held 0.845 and 0.701: short at long horizons too, but far closer. A likely reason, not tested here: the curve's misses come in runs, and an interval built from independent monthly noise does not allow for them.

WHY CHOOSE THIS?

Refitting is nearly free, so the curve can always use the latest month.

WHY NOT THIS? (TRADEOFFS)

Even refitted, it ignores how far off the curve the last months were, unless something like the AR(1) is added.

6. Electrical-equipment orders, and where the slope changed

DEFINITION — start here, no prior knowledge assumed

On the log orders the settings are chosen on 49 validation origins before each cut: 2008-01 (the crash) and the last 24 months. Then the whole series is fitted once, as a description rather than a forecast, to see where the slope changed most.

REAL-WORLD SCENARIO

A sales chart reviewed after a recession: the model cannot have predicted the drop, but it can date it.

ACTUAL OUTPUT

```
the crash: train up to 2007-12; validation chose 5 waves, penalty 0.01; 24 months, MAE:
  Prophet-style 29.03%; + AR(1) 29.01%; line + season 25.99%; airline 23.86%; Holt-Winters 18.81%; regression 18.41%; LSTM
      avg 25.57%; control 23.06%; last year 18.04%
the last 24 months: train up to 2014-05; validation chose 5 waves, penalty 1000; 24 months, MAE:
  Prophet-style 5.86%; + AR(1) 5.77%; line + season 12.90%; airline 6.41%; Holt-Winters 4.65%; regression 2.72%; LSTM avg
      5.30%; control 2.37%; last year 3.75%
the whole series (4 waves, penalty 1, changepoints to 2012-02): largest slope changes 2009-05 +0.180, 2008-01 -0.129,
    2010-01 +0.117 (log units a year)
Actual output from this lesson's run.
```

What it means

Through the crash validation chose a light penalty (0.01), and the curve scored 29.03%, the worst of all; Holt-Winters (18.81%), the regression (18.41%) and last year's value (18.04%) did best. Over the last two years it scored 5.86%, beating the airline model (6.41%) but not the control (2.37%) or the regression (2.72%).

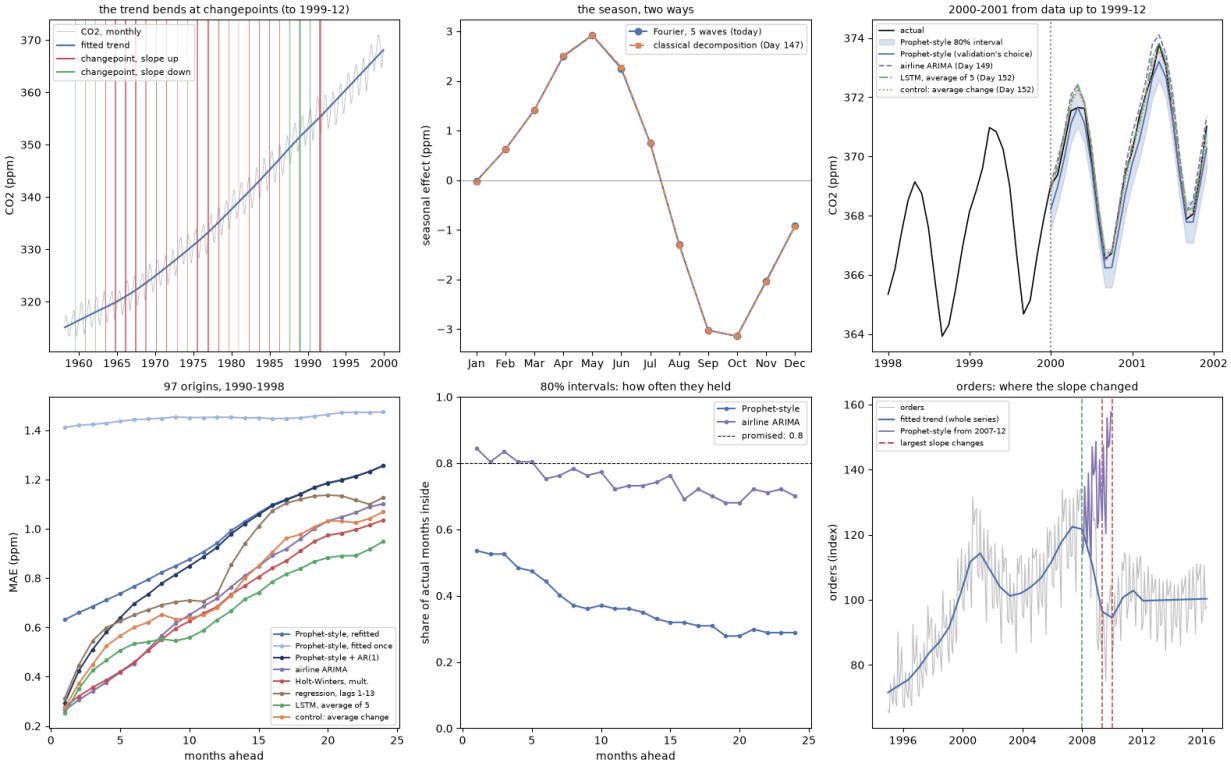
Looking back over the whole series, the largest slope changes fell at 2008-01 (-0.129) and 2009-05 (+0.180): the start of the crash and the recovery, found without being told.

WHY CHOOSE THIS?

Changepoints date the turns in a series: a useful description after the fact.

WHY NOT THIS? (TRADEOFFS)

A trend fitted to a boom carries the boom's slope into the bust; no curve of time foresees a turn.



Top: CO2's fitted trend with its changepoints (line width: size of the slope change); the season against Day 147's; the 2000–2001 forecasts with the 80% interval. Bottom: rolling-origin error by horizon for every model; how often the 80% intervals held; the orders with the fitted trend, its largest slope changes and the forecast from 2007-12.

Interview preparation — technical questions

Q: How does a Prophet-style model represent the trend?

A: A line plus a hinge (t - c)+ at each candidate changepoint; each hinge's coefficient is a change of slope, penalised so most stay small.

Q: How do you fit it without a special solver?

A: With a squared penalty, it is least squares on the design matrix stacked over sqrt(lambda) rows: here it matched the closed form to 1.9e-11.

Q: How should the penalty be chosen?

A: On time-ordered validation: here it chose 100 at 0.474, which then ranked 5 of 6 on the test.

Q: How do you check a forecaster's intervals?

A: Count how often actual values fall inside over many origins: 80% intervals held 0.536 to 0.289 here, the airline model's 0.845 to 0.701.

Q: How do you test why a model loses at short horizons?

A: Measure the residuals' memory (lag-1 autocorrelation 0.819) and add it back: an AR(1) cut the one-month error from 0.631 to 0.294.

Interview preparation — theoretical questions

Q: What does a curve-of-time model give up?

A: It forecasts from the date, not from the last values, so it ignores how far off the curve the series currently is.

Q: Why are its intervals too narrow here?

A: They treat months as independent noise around the curve, but the residuals came in long runs (Ljung-Box 1633.6).

Q: Where does it shine?

A: Description: a season matching Day 147's to 0.028 ppm, readable slopes, and turns dated after the fact (2008-01, 2009-05).

Q: Why place changepoints only in the first 80%?

A: So the last slope is estimated from a long stretch; a bend fitted to the last few months would be carried forever.

Key takeaway and what's next

> **KEY TAKEAWAY**
>
> The week's parts added up to a Prophet-style forecaster: hinges and waves, one penalised least squares, simulated intervals. It reads well: slopes in ppm a year, a season matching Day 147's, slope changes dated at the 2008 crash and recovery. As a forecaster it was middling on 2000–2001, lost to the airline ARIMA at every horizon over 97 origins, and its intervals held far less often than promised. The cause was measured: its residuals have strong memory. Adding that memory back with an AR(1) more than halved its one-month error. Week 22's models, side by side, show no single winner: over the rolling origins the LSTM led at 1, 12 and 24 months, the airline model at 6, and the simple control beat the curve at every horizon shown.

- Checks: slopes 0.937, 3.058, 0.449 for 1, 3, 0.5; least squares agrees to 2.1e-13.
- Validation chose penalty 100; it ranked 5 of 6 on the test.
- CO2 2000–2001: curve 0.400, plus AR(1) 0.295, airline 0.412, control 0.231.
- Rolling: beat the airline model at 0 of 24 horizons; 80% intervals held 0.536 to 0.289.
- Orders: largest slope changes at 2008-01 and 2009-05.

Suggested watch order, if starting from zero

- 1. Sean Taylor - Forecasting at Scale: How and Why We Developed Prophet for Forecasting at Facebook — the talk by one of Prophet's authors on how and why it was built, with its changepoints and intervals.
- 2. Forecasting with the FB Prophet Model (Rob Mulla) — Prophet used in Python on a real series, to set beside today's from-scratch version.

Day 154 preview

- **Week 23 begins: recommender systems.** Day 154 starts with collaborative filtering, user-based and item-based, on a real ratings matrix.